Utility functions, one form at a time

Lecture notes

What this lecture is about. Last lecture ended with a picture: the best feasible bundle is where the highest reachable indifference curve touches the budget line. Today we go slowly through the utility functions, one form at a time: linear, Leontief, Cobb-Douglas and quasi-linear, and what their parameters do to the indifference curves. Finding the best bundle with each of them continues in Lecture 10.

One budget. Prices $p_x = 3$, $p_y = 1$, money $B = 60$:

$$3x + y = 60.$$

1 Utility functions, one form at a time

Utility function

A utility function gives every bundle a number, so that a better bundle gets a higher number and equally good bundles get the same number. An indifference curve is all bundles with the same number. Only the order of the numbers matters (ordinal): $2u + 5$ or u^3 describe the same preferences.

form	formula	indifference curves
linear (perfect substitutes)	$u = a\,x + b\,y$	straight lines; the MRS is a/b everywhere
Leontief, $a = b = 1$ (perfect complements)	$u = \min\{x, y\}$	L-shapes, kinks on the line $x = y$
Leontief (the general recipe)	$u = \min\{a\,x, b\,y\}$	L-shapes, kinks on the line $a\,x = b\,y$
Cobb-Douglas	$u = x\,y$ (or $x^\alpha y^\beta$)	smooth, bowed in
quasi-linear	$u = \alpha x + y^\beta$, e.g. $2x + \sqrt{y}$	each curve is the same curve shifted; bowed in for $\beta < 1$
quasi-linear, y squared	$u = x + y^2/60$	shifted copies, but bowed *out*

Perfect complements are the 1 : 1 Leontief ($a = b = 1$). Example of a recipe: every coffee needs exactly 2 sugars, $u = \min\{2x, y\}$, with kinks on $y = 2x$.

A number for every bundle. With $u = x\,y$: $(2, 8)$, $(4, 4)$ and $(8, 2)$ all get 16, so they lie on one indifference curve; $(4, 8)$ and $(8, 4)$ get 32, on a higher one. Double every number, $v = 2xy$: 16 becomes 32 and 32 becomes 64, but the curves do not move and the order stays. Only the order matters: utility is *ordinal*.

Same bundle, different preferences. On the parameter sliders every curve goes through the same bundle $A = (10, 10)$: each curve is a *different person*. One person's indifference curves never cross; different people's can. Ask each person: one more x, how much y would you give? That is the MRS at A:

$$\text{substitutes: } \frac{a}{b}, \qquad \text{Cobb-Douglas: } \frac{\alpha}{\beta} \cdot \frac{y}{x}, \qquad \text{quasi-linear: } \frac{\alpha}{\beta} y^{1-\beta}.$$

For perfect complements: on the flat arm nothing (extra x alone adds nothing), on the steep arm the spare y down to the kink. A steeper curve through A means a person who likes x more.

Quasi-linear, the other way: y squared

With $u = x + y^2/60$ (the $1/60$ only makes the picture visible), $MU_x = 1$ and $MU_y = y/30$, so $MRS = 30/y$: the curves are again shifted copies, but now they bow *out*, because more y makes y worth more, not less. On the budget $3x + y = 60$:

bundle	$(20, 0)$	$(50/3, 10)$	$(10, 30)$	$(0, 60)$
u	20	18.33	25	**60**

The curve just touches the line at $y = 10$ (where $30/y = 3$), and that point is the *worst* on the line. The best is the corner $(0, 60)$. Touching is not enough when the curves bow out.

The general idea: y enters through a function. Quasi-linear means $u = \alpha x + v(y)$: perfect substitutes, but y enters through a function $v(y) = y^\beta$. With $\beta = 1$ it is plain perfect substitutes; with $\beta < 1$ each extra y adds less (bowed in, as with the coffees); with $\beta > 1$ each extra y adds *more*. Think of craving (the more you've had, the more you want the next one), binge-watching (the more episodes you have seen, the more you want the next one). The curves bow out, so the best bundle is a corner: all or nothing. (Real addiction runs over time, yesterday to today; Frank ch. 7 shows the other side, tolerance and self-control, pp. 231–242.) **Next: Lecture 10.** Marginal utility and the MRS from a utility function, then the best feasible bundle for each form: at a corner, at a kink, or where a curve just touches the line.

Reading. Frank, *Microeconomics and Behavior*, pp. 71–76 and Appendix 3A, pp. 87–92 (the utility mountain: Fig. 3A.3). Online: M. J. Osborne, *Mathematical Methods for Economic Theory* (free tutorial), §2.3 (the gradient stands at a right angle to the level curve) and §6.1 (one constraint; what λ means), `https://mjo.osborne.economics.utoronto.ca/index.php/tutorial`.